

Project Design Phase-II

Solution Requirements (Functional & Non-Functional)

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Team ID: VIP-C2-Book-A-Doctor-MERN

Project Name: Book A Doctor – Online Doctor Appointment Booking System

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Registration Form
		User Profile Creation
FR-2	User Authentication	Login using Email and Password
		Logout Functionality
FR-3	Doctor Management	Apply for Doctor Account
		Admin Approval of Doctor Applications
FR-4	Doctor Search	View Approved Doctors
		Search Doctor Details
FR-5	Appointment Management	Book Appointment
		View Appointment Status
		Cancel Appointment
FR-6	Doctor Dashboard	View Appointment Requests
		Approve/Reject Appointments
FR-7	Notification Management	Receive Appointment Notifications

FR-8	Admin Management	Receive Doctor Approval Notifications
		Manage Users
		Manage Doctors
		Monitor System Activities

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide a simple, user-friendly, and responsive interface for patients, doctors, and administrators.
NFR-2	Security	User passwords must be encrypted using bcryptjs, and secure authentication must be implemented using JWT.
NFR-3	Reliability	The application should consistently perform operations without data loss or system failures.
NFR-4	Performance	System responses should be fast, and appointment operations should be processed efficiently.
NFR-5	Availability	The application should be accessible online with minimal downtime.
NFR-6	Scalability	The system should support increasing numbers of users, doctors, and appointments without significant performance degradation.

Conclusion

The functional requirements define the core features required for the Book A Doctor platform, while the non-functional requirements ensure that the system remains secure, reliable, scalable, and user-friendly. Together, these requirements provide a strong foundation for successful system development and deployment.